

CS & IT ENGINEERING

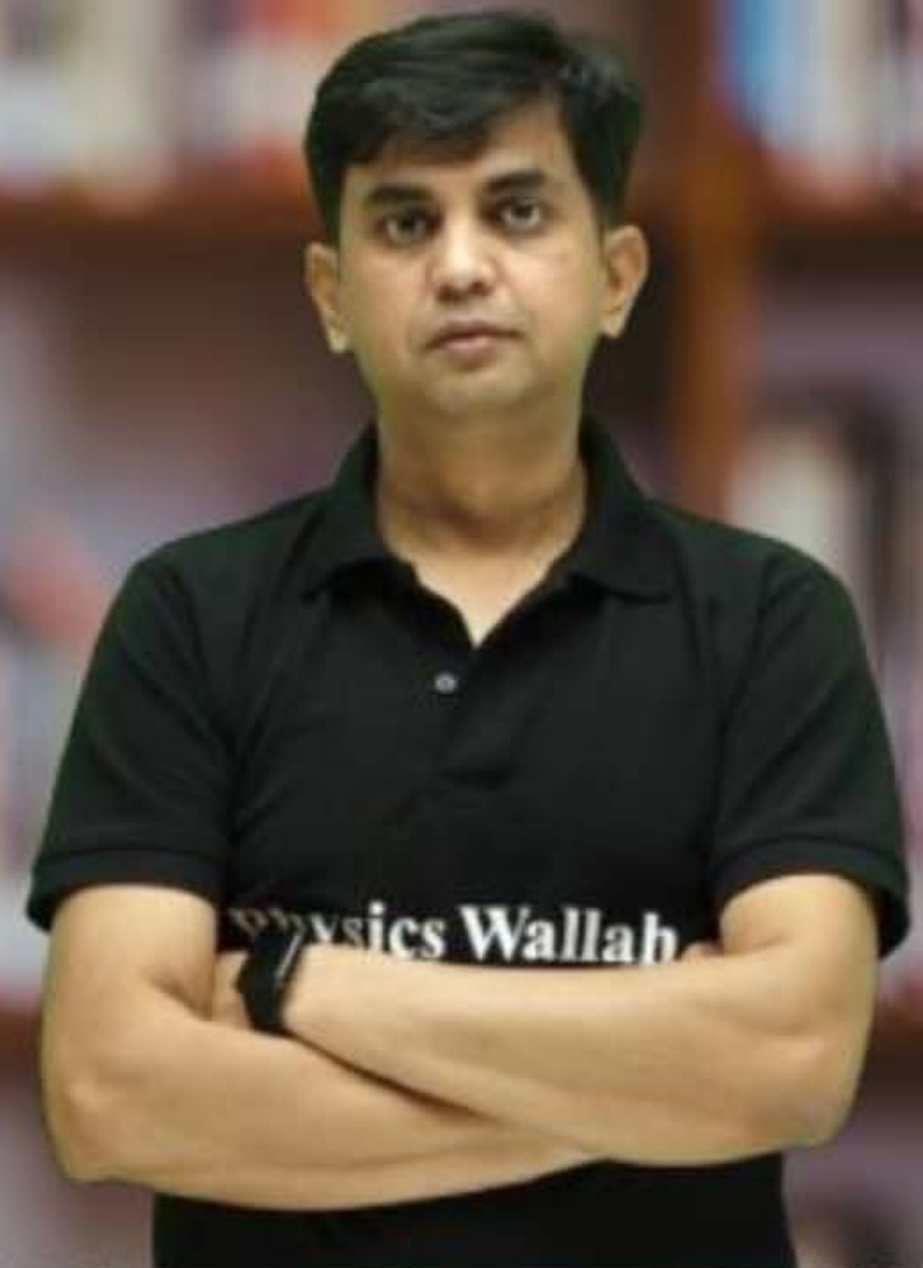


Computer Network

Routing

Lecture No. - 02

By - Abhishek Sir





Recap of Previous Lecture



Topic

Routing

Topic

Static Routing



- ① Shortest path Algo.
- ② Flooding



Topics to be Covered



Dynamic Routing



Topic

Link State Routing

Topic

Distance Vector Routing



ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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[MCQ]

[GATE-2025] [1 Mark]



#Q. Consider the routing protocols given in List I and the names given in List II:

List I	List II
(i) Distance vector routing	(a) Bellman-Ford
(ii) Link state routing	(b) Dijkstra

For matching of items in List I with those in List II, which ONE of the following options is CORRECT?

- ☒ A (i) – (a) and (ii) – (b)
- ☐ B (i) – (a) and (ii) – (a)
- ☐ C (i) – (b) and (ii) – (a)
- ☐ D (i) – (b) and (ii) – (b)

Ans: A

[MCQ]

(IGP) [GATE-2014] [1 Mark]



#Q. Which one of the following is TRUE about interior Gateway routing protocols - Routing Information Protocol (RIP) and Open Shortest Path First (OSPF) ?

- ☒ A RIP uses distance vector routing and OSPF uses link state routing
- ☐ B OSPF uses distance vector routing and RIP uses link state routing
- ☐ C Both RIP and OSPF use link state routing
- ☐ D Both RIP and OSPF use distance vector routing

(RIP-DV, OSPF-LS)

Ans: A



Topic : Link State Routing

→ Centralized Routing Algorithm

[determine least-cost path using complete picture of the network topology]

→ Link State Routing is divided into three steps :

1. Maintain Link State information

2. Link State broadcast

3. Share updated Link State information

→ Initialization



Topic : Link State Routing

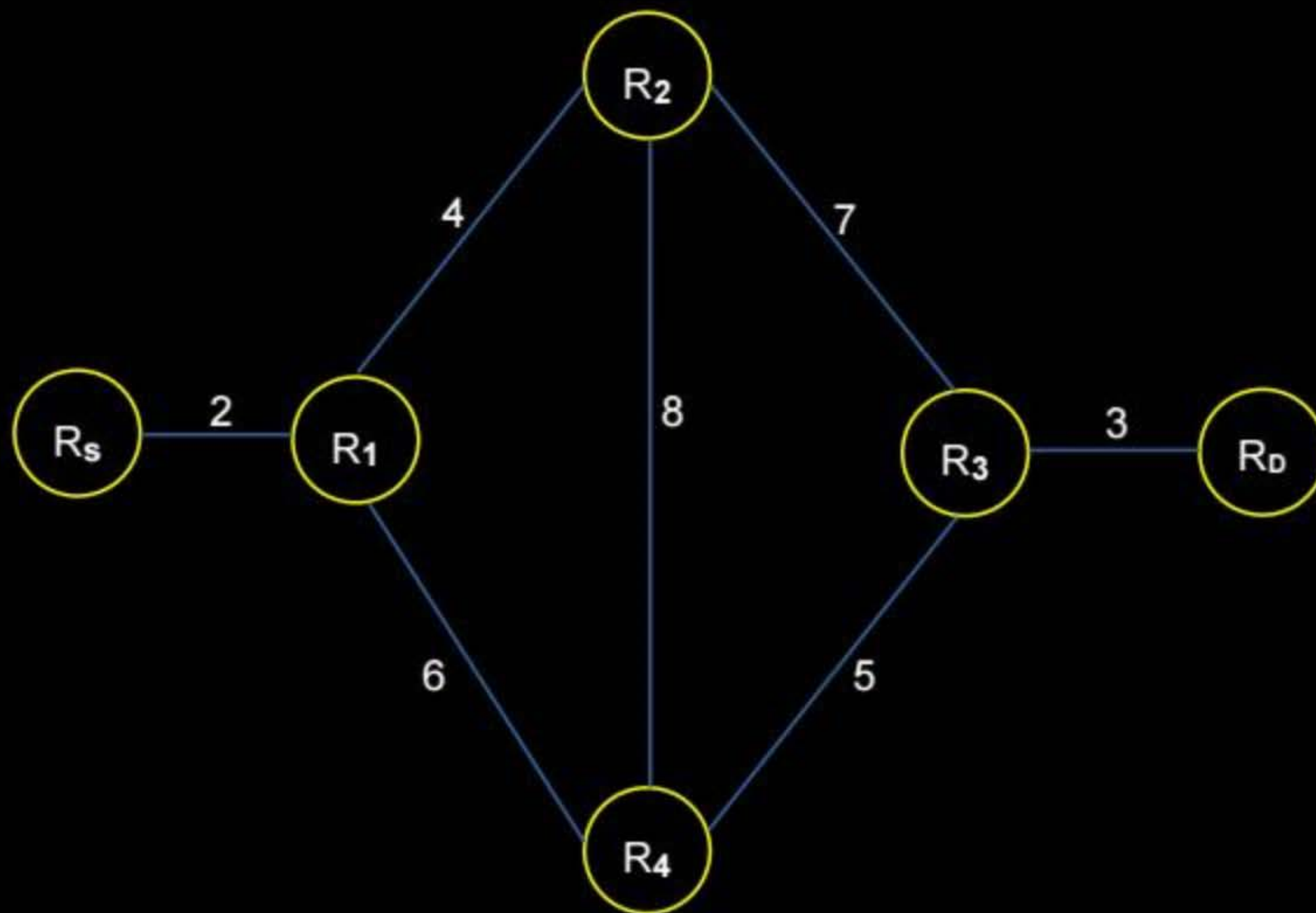


1. Maintain Link State information

- Each router maintain separate "Link State" information
- Maintain information about adjacent link only
[Adjacent (neighbor) routers only]



Topic : Link State Routing





Topic : Link State Routing



R_s Link State Info

R ₁	2
----------------	---

R_D Link State Info

R ₃	3
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R₁ Link State Info

R _s	2
R ₂	4
R ₄	6

R₂ Link State Info

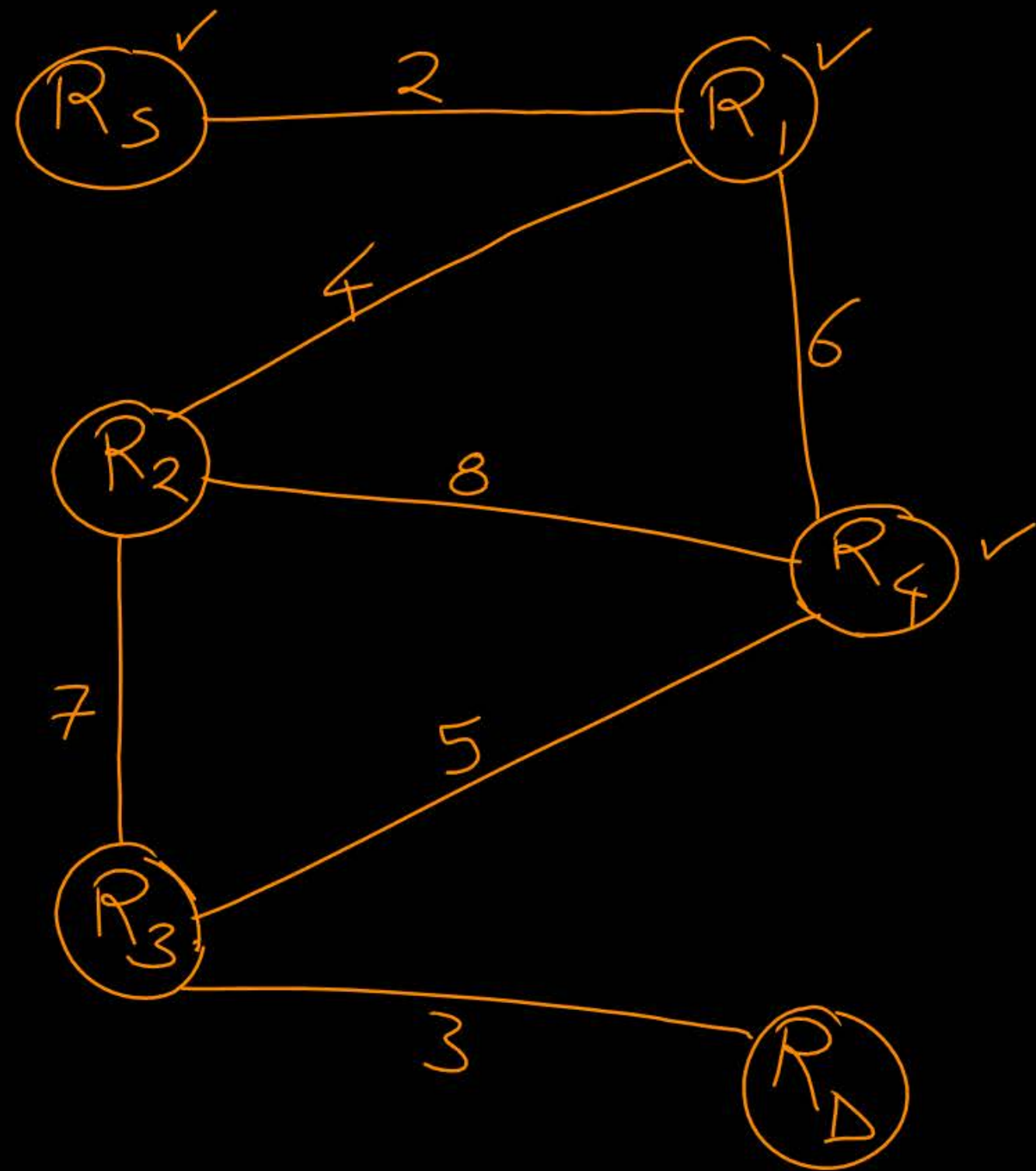
R ₁	4
R ₃	7
R ₄	8

R₃ Link State Info

R _D	3
R ₂	7
R ₄	5

R₄ Link State Info

R ₁	6
R ₂	8
R ₃	5



Isomorphic graph 



Topic : Link State Routing

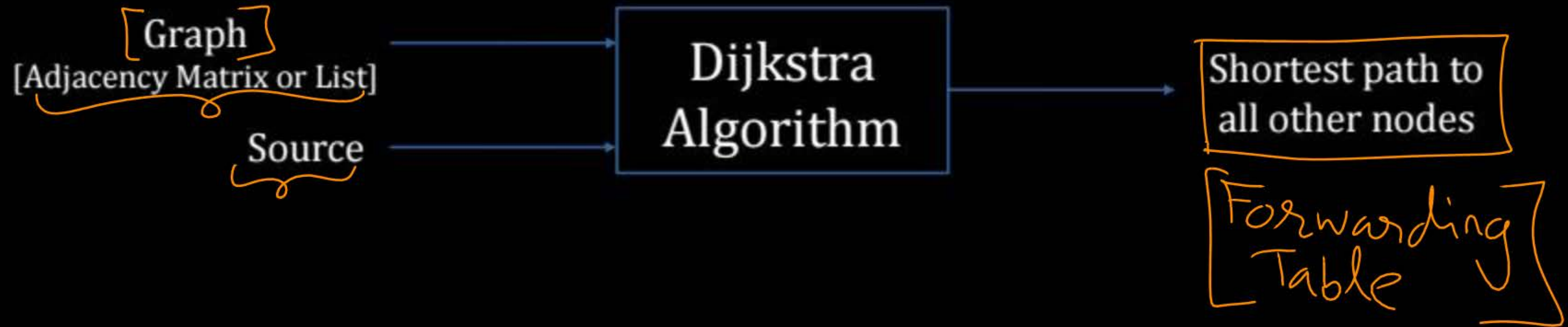


2. Link State broadcast

- Each router flood (broadcast) its "Link State" information *in the network*
[To all other routers in the network]
- Each router have "Link State" information of all other routers
[Complete information about the network topology]
- Each router construct "Adjacency Matrix" or "Adjacency List"
- Each router execute "Dijkstra's Algorithm" locally
[To find optimal path to all other routers]
(Assum itself as a source)



Topic : Link State Routing





Topic : Link State Routing



* Asynchronous Update

3. Share updated Link State information

- Whenever changes occur in the network topology,
[Relevant routers update their "Link State" information accordingly]
- Only those routers broadcast their updated "Link State" information
[To all other routers in the network]
- Whenever a router receive any updated "Link State" information,
it construct new "Adjacency Matrix / List" accordingly and
re-execute "Dijkstra's Algorithm" locally



Topic : Dijkstra's Algorithm



Dijkstra's Algorithm time complexity

- Number of nodes = n
- Total n iteration
- Total number of comparisons = $O(n^2)$

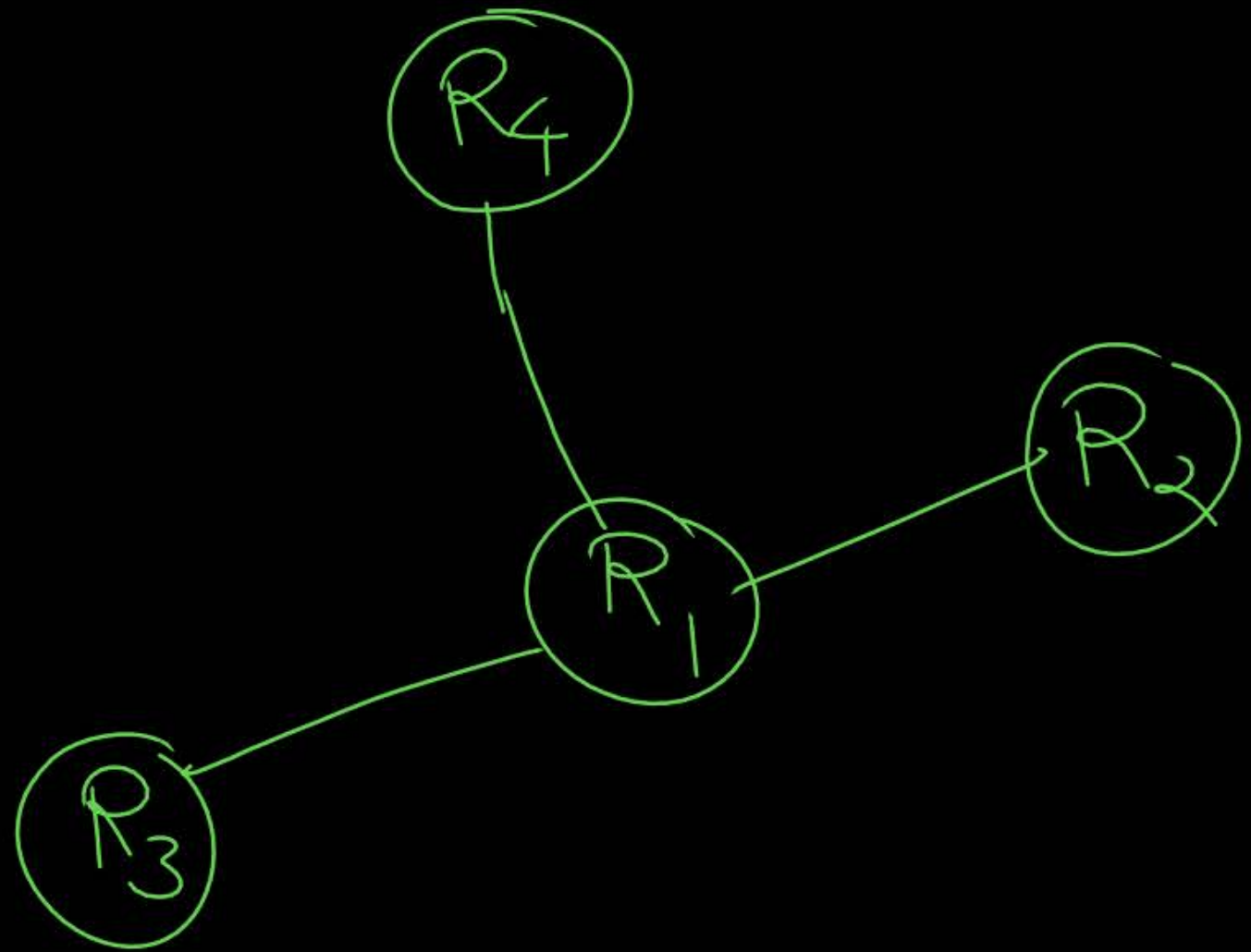
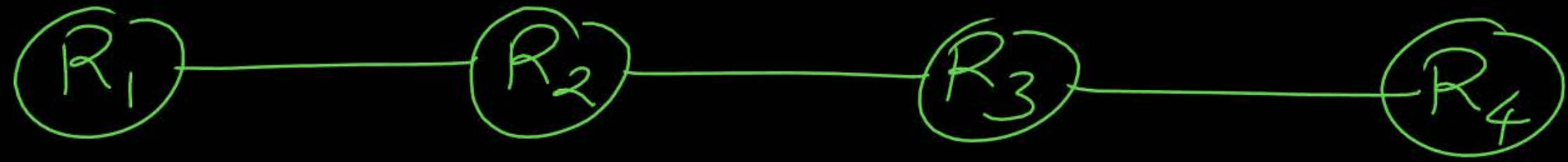


Topic : Link State Routing



Message Complexity :

- Number of nodes = n
- Each router must broadcast its Link State information to all other router
- Each router's message crosses $O(n)$ links
- Overall message complexity = $O(n^2)$ ✓



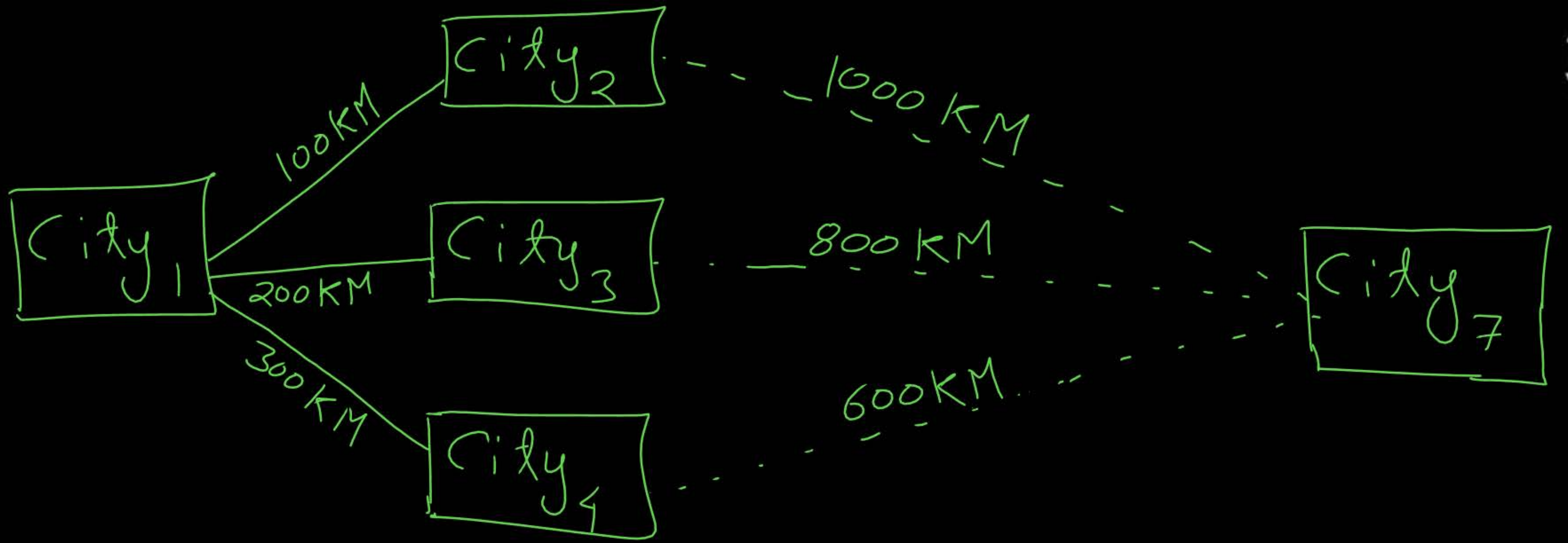


Topic : Link State Routing

[w.r.t. DV Routing]



- High number of packets per network $\Rightarrow$ Dis Adv.
- Adapt network topology very fast $\Rightarrow$ Adv.



$$D_1(7) =$$



Topic : Distance Vector Routing

→ Based on Bellman-Ford equation
[Dynamic programming]

→ Bellman-Ford equation

Let $D_x(Y)$: Cost of the least-cost path from X to Y

$$D_x(y) = \min_v \{ C(X, V) + D_v(Y) \}$$

→ Minimum taken over all neighbors V of X

$C(X, V)$: Direct cost of link from X to V

$D_v(Y)$: V's estimated least-cost path cost to Y



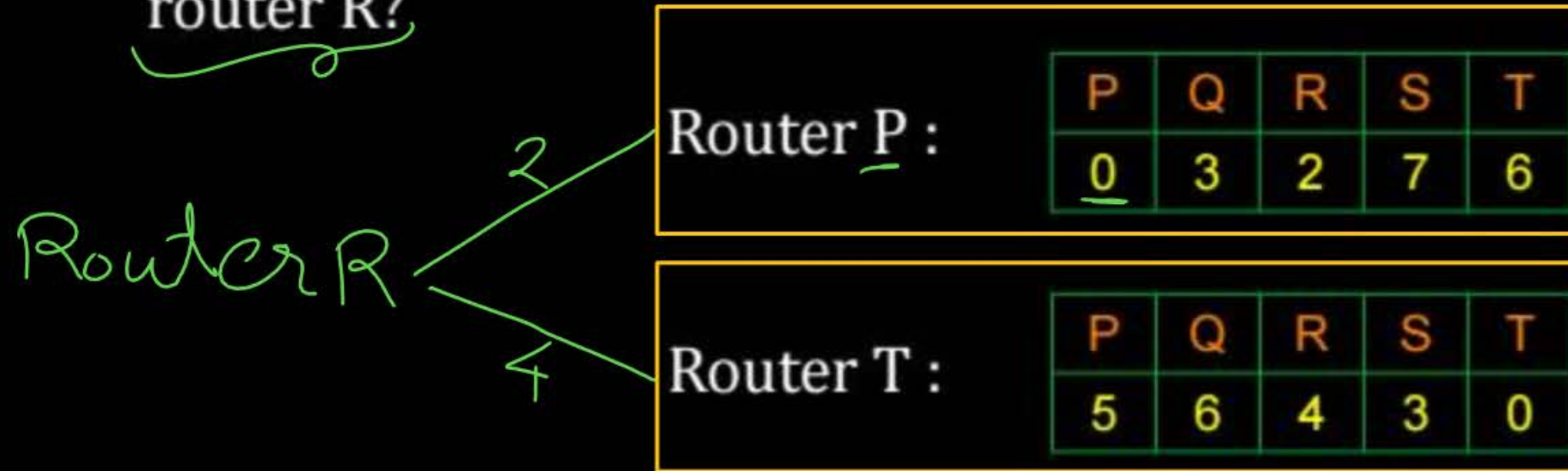
$$D_x(Y) = \min [C(X, V_1) + D_{V_1}(Y), \\ C(X, V_2) + D_{V_2}(Y), \\ C(X, V_3) + D_{V_3}(Y)]$$



Topic : Distance Vector Routing

- Each router maintain separate “Distance Vector” estimate
[Best known minimum distance to all other routers]
- Each router sends their own “Distance Vector” estimate to
their neighbor routers only
- When a router receives new “Distance Vector” estimate from any neighbor,
It update its own “Distance Vector” using Bellman-Ford equation

#Q. Consider network of five routers (P, Q, R, S and T). Router P and T are direct neighbour of router R with distance 2 and 4 respectively. Consider following is the distance vector of router P and T then calculate distance vector of router R?



Router R :	Node	P	Q	R	S	T
	Cost	2	5	0	7	4
	Via	P	P	R	T	T

#Q. For the network given in the figure below, the routing tables of the four nodes A, E, D and G are shown. Suppose that F has estimated its delay to its neighbors, A, E, D and G as 8, 10, 12 and 6 msec respectively and updates its routing table using distance vector routing technique.

Routing Table of A	
A	0
B	40
C	14
D	17
E	21
F	9
G	24

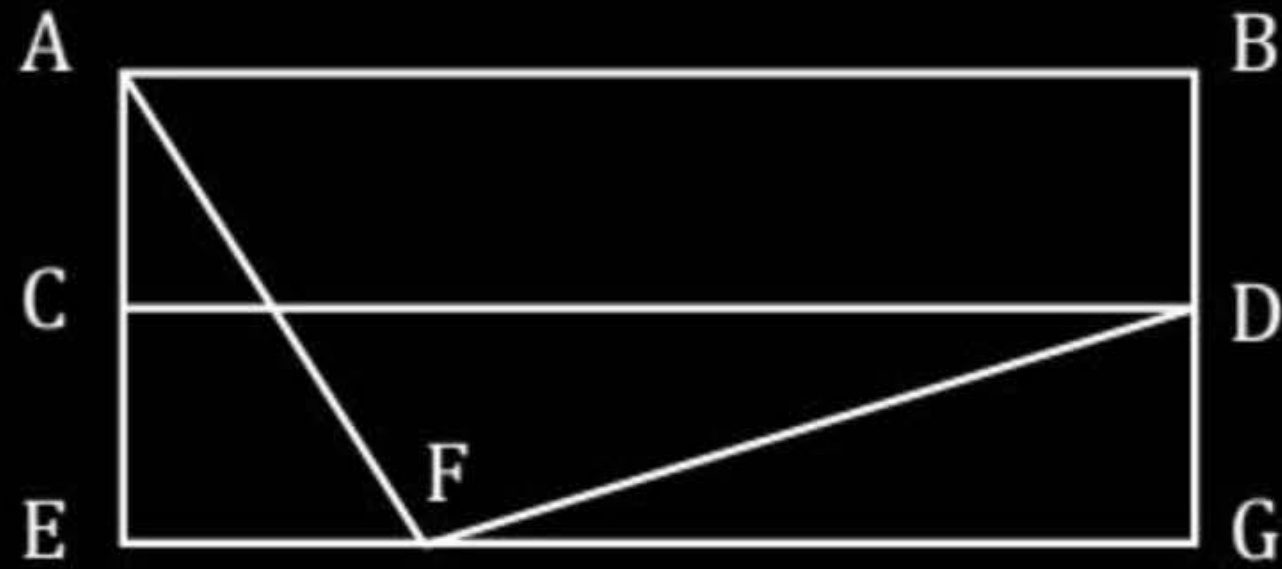
Routing Table of D	
A	20
B	8
C	30
D	0
E	14
F	7
G	22

Routing Table of E	
A	24
B	27
C	7
D	20
E	0
F	11
G	22

Routing Table of G	
A	21
B	24
C	22
D	19
E	22
F	10
G	0

[GATE-2007]

*11T-1R
H.W.*


A

A	8
B	20
C	17
D	12
E	10
F	0
G	6

B

A	21
B	8
C	7
D	19
E	14
F	0
G	22

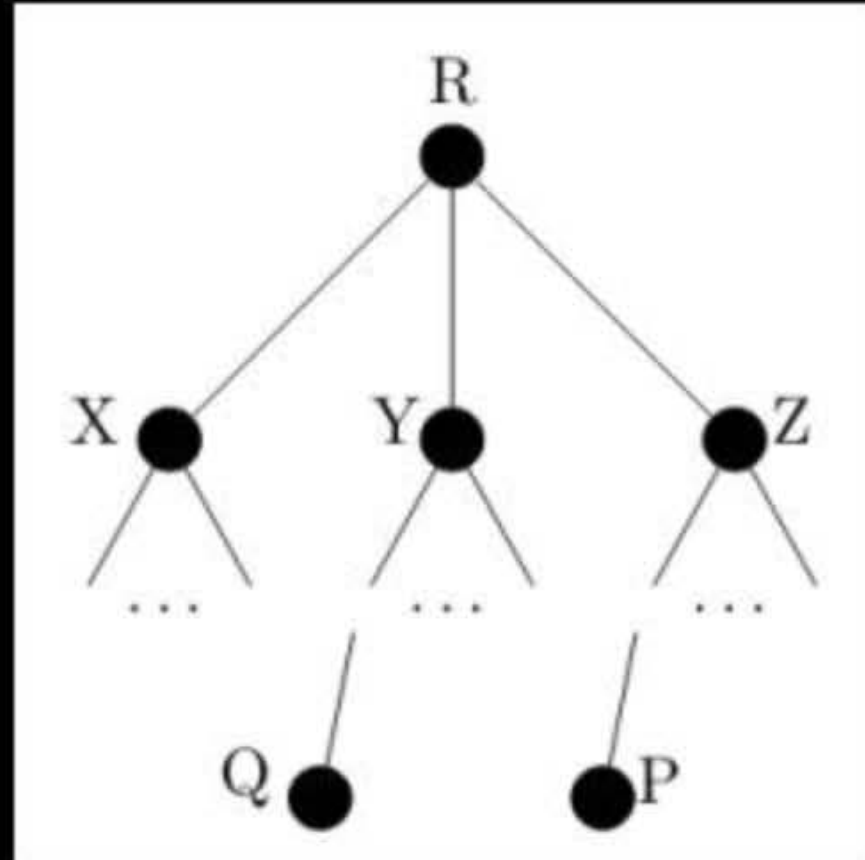
C

A	8
B	20
C	17
D	12
E	10
F	16
G	6

D

A	8
B	8
C	7
D	12
E	10
F	0
G	6

#Q. Consider a computer network using the distance vector routing algorithm in its network layer. The partial topology of the network is shown below.



The objective is to find the shortest-cost path from the router R to routers P and Q. Assume that R does not initially know the shortest routes to P and Q. Assume that R has three neighbouring routers denoted as X, Y and Z.

During one iteration, R measures its distance to its neighbours X, Y, and Z as 3, 2 and 5, respectively. Router R gets routing vectors from its neighbours that indicate that the distance to router P from routers X, Y and Z are 7, 6 and 5, respectively. The routing vector also indicates that the distance to router Q from routers X, Y and Z are 4, 6 and 8 respectively. Which of the following statement(s) is/are correct with respect to the new routing table of R, after updation during this iteration?

- (A) The distance from R to P will be stored as 10
- (B) The distance from R to Q will be stored as 7
- (C) The next hop router for a packet from R to P is Y
- (D) The next hop router for a packet from R to Q is Z

[GATE-2021]

11T-B
(H.W)



2 mins Summary



Topic

Link State Routing ✓

Topic

Distance Vector Routing



THANK - YOU

